MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Multi-Disciplinary Courses (MDC) - MG1MDCCFS101 - DATA RECOVERY

Duration: 1.5 Hours Maximum Marks: 35

Part A : Answer **any 10** Questions

Each Question carries **1 mark**

1. Data from a formatted hard disk can be recovered using specialized recovery software if the disk hasn't been overwritten with new data. Specify any recovery software tool with one sentence (Disk-Drill, R-studio, Easeus, Recuva etc) (Explanation : 0.5 Mark , Mention software name : 0.5 Mark)
2. Explain any two type of data loss: i.e, Physical data loss, logical dataloss or accidential deletion, physical damage,hardware or software loss ,natural disaster ,Theft , cyber attacks etc) (**0.5 marks each** any two types of data loss)
3. Live imaging refers to creating a complete image (bit-for-bit copy) of a storage device while the system remains running and operational. it allows analysts to capture all data, including deleted and hidden files, without shutting down or disrupting system activities. (Give 0.5 mark if they explain disk imaging)
4. Software issues causing data loss include software corruption or crashes, errors during file saving or backup, antivirus programs mistakenly deleting files, and malware or viruses damaging or deleting data. These issues can lead to files becoming corrupted, lost, or inaccessible .
5. A ransomware attack is a type of malware that encrypts a victim's data and demands a ransom payment to restore access to the files, effectively holding the data hostage until the payment is made or the data is otherwise recovered.

6. A keylogger is a hidden program or device that records everything you type on your keyboard, such as passwords and messages, without you knowing, and sends this information to someone who wants to steal it. (give 0.5 mark if they explain malware)
7. A full disk image is an exact, complete copy of all the data and structure on a hard drive or storage device, including the operating system, files, settings, and even hidden or deleted data, saved as a single file to be used for backup or recovery purposes.
8. The price of a data recovery tool matters because it helps to choose one that fits the budget while still being good enough to recover the lost files safely and effectively. It's important to find a balance between cost and the tool's features to get the best value without overspending.
9. Rotational latency is the time it takes for a hard drive's spinning disk to rotate so the required data sector is under the read/write head, usually measured in milliseconds, and depends on the disk's rotational speed (RPM). (Give 0.5 mark if they mention platter/spindle)
10. The most efficient backup strategy for minimizing storage is incremental backup, which saves only the changes made since the last backup, using less space and making backups faster . (Mention the backup strategy name : **0.5 Mark**, define : **0.5Mark**)
11. The full form of RAID is **Redundant Array of Independent Disks.** It is a technology that combines multiple physical disk drives into one logical unit to improve data reliability, performance, and storage capacity through techniques like data striping, mirroring, and parity. (Full form : **1 Mark**)
12. Explain any two causes of file system corruption : Sudden shutdowns or power loss during disk operations, Faulty storage devices, such as drives with bad sectors or hardware failures, leading to damaged or unreadable file system areas. (Give **0.5 mark** for any two causes)

Part B : Answer **any 3** Questions

Each Question **carries 5 mark**

13. Explain disk partitioning : Disk partitioning is the process of dividing a single physical hard drive into multiple separate sections called partitions (**2 Marks**)

    Explain its steps for creating a new disk partition (ie creating new volume disk ,local disk F,E etc)(**2 Marks**)

    Explain the mounting of disk partitioning (**Explanation of mounting :0.5 Mark, Example: 0.5 mark)**

14. Explain any 3 list of data loss (i.e., Accidental deletion , File system corruption ,formatting errors , cyber attacks, natural disasters and environmental events ,hardware and software failure etc.)

    (Define the data loss **: 0.5 marks ,** Explain **any 3 lists** of data loss**: 1marks each ,** Example **: 0.5 marks each**)

15. Explain any 4 types of memory cards (i.e. 1.SD (Secure Digital) card: A popular memory card used for storing data in devices like cameras and phones, available in various capacities and speed classes.

    2. microSD card: A smaller version of the SD card, commonly used in smartphones and compact devices, offering similar functionalities.

    3. CompactFlash (CF/CFast): High-performance memory cards mainly used in professional cameras, with CFast offering faster SATA-based speeds.

    4. XD-Picture/SmartMedia/MMC: Older, less common memory cards once used in cameras and early digital devices, now largely obsolete.

    5. Memory Stick (MS) Card: A proprietary Sony format used primarily in Sony devices like cameras and PlayStation consoles.)

    (Define use of memory card **: 1 marks ,** Explain **any 4 types** of memory card**: 1marks each**)

16. Explain malware with any 3 types of attacks : Malware, short for malicious software, is any software intentionally designed to damage, disrupt, or gain unauthorized access to computer systems, networks, or data, often compromising security and privacy. i.e,
    1. Virus**:** A malware that attaches to files and spreads by infecting other files, causing damage and corruption.
    2. Worm**:** A self-replicating malware that spreads across networks without user interaction, often consuming bandwidth.
    3. Trojan Horse**:** Malicious software disguised as legitimate, which creates backdoors for unauthorized access.
    4. Ransomware**:** Malware that encrypts files and demands payment to restore access.
    5. Spyware**:** Software that secretly monitors user activity and steals sensitive information.

    Each type of attack has distinct methods to harm or control computer systems.

    (Define malware : **0.5 marks** , Explain **any 3 types** of attacks : **1 marks each ,** Example **: 0.5 Marks each)**

Part C : Answer **any 1** Question

Each Question **carries 10** marks

17. A file system is a method used by an operating system to organize, store, retrieve, and manage files on a storage device, enabling efficient access to data by maintaining a structured hierarchy of files and directories along with metadata for each file.

    Types : NTFS supports large file sizes, file permissions, encryption, and journaling for security and reliability; FAT32 is compatible with most devices but limited to 4GB file size and lacks advanced features; exFAT combines FAT32's compatibility with support for larger file sizes and is optimized for flash drives but lacks NTFS's security features.

(Define file system : **2 Marks** , Explain **any 4 differences** of each types : **1 marks each** , Example of each types : **1 marks each**)

18. Data recovery is the process of restoring lost, deleted, corrupted, or inaccessible data from storage devices such as hard drives, SSDs, USB drives, or memory cards using specialized software or techniques to make the data usable again.

Tools : **Recuva:** A user-friendly Windows tool for recovering deleted files with quick and deep scans, supporting common file types and FAT, exFAT, and NTFS file systems. **Recoverit:** A versatile data recovery software capable of restoring lost files from various devices with advanced scanning and supported formats. **EaseUS Data Recovery:** A comprehensive recovery program that supports multiple file types and devices, featuring quick scanning and preview options. **Disk Drill:** A powerful and user-friendly tool that recovers lost data from various file systems and devices with advanced scanning algorithms. **R-Studio:** An advanced recovery software favored by IT professionals for recovering data from complex file systems and damaged RAID configurations. **PhotoRec:** An open-source data recovery tool focusing on recovering lost files from damaged or formatted storage, particularly effective for photos. **Pandora Recovery:** Offers simple and effective file recovery on Windows with a focus on deleted file recovery and file preview capabilities.

(Define data recovery : **1 Marks ,** Explain **any 2 tools** : **4 marks each** , Example of two types :**0.5 marks each**)